

Translated

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Dijkstra Algorithm

You are given a directed or undirected weighted graph with n vertices and m edges. The weights of all edges are non-negative. You are also given a starting vertex s . This article discusses finding the lengths of the shortest paths from a starting vertex s to all other vertices, and output the shortest paths themselves.

This problem is also called **single-source shortest paths problem**.

Algorithm

Here is an algorithm described by the Dutch computer scientist Edsger W. Dijkstra in 1959.

Let's create an array $d[]$ where for each vertex v we store the current length of the shortest path from s to v in $d[v]$. Initially $d[s] = 0$, and for all other vertices this length equals infinity. In the implementation a sufficiently large number (which is guaranteed to be greater than any possible path length) is chosen as infinity.

$$d[v] = \infty, \forall v \neq s$$

In addition, we maintain a Boolean array $u[]$ which stores for each vertex v whether it's marked. Initially all vertices are unmarked:

$$u[v] = \text{false}$$

The Dijkstra's algorithm runs for n iterations. At each iteration a vertex v is chosen as unmarked vertex which has the least value $d[v]$:

Evidently, in the first iteration the starting vertex s will be selected.

The selected vertex v is marked. Next, from vertex v **relaxations** are performed: all edges of the form (v, to) are considered, and for each vertex to the algorithm tries to improve the value $d[\text{to}]$. If the length of the current edge equals len , the code for relaxation is:

$$d[\text{to}] = \min(d[\text{to}], d[v] + len)$$